Functional Requirements Specification (FRS)

Phase 2: System Capabilities & Behavioral Specifications

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1. Core Functional Requirements Matrix

The system must execute the following functional behaviors reliably and deterministically across all environments.

- FR-01 (Data Cleaning): The system shall ingest raw CSV datasets, impute missing geographic data via NOC lookups, and standardize medal values to ['Gold', 'Silver', 'Bronze', 'No Medal'].
- FR-02 (Deduplication): The system shall aggregate team event medals such that each national team victory contributes exactly one medal to the country's national total.
- FR-03 (Weighted Scoring): The system shall calculate a weighted score for each country-year tuple using the formula: Weighted Score = 3*Gold + 2*Silver + 1*Bronze.
- FR-04 (Host Surge Tracking): The system shall compute pre-games baseline medals, host-year medals, and net bump for all Olympic host nations.
- FR-05 (REST API): The Flask web server shall expose endpoints /api/kpis, /api/medals, and /api/host-surge returning JSON data with HTTP 200 status.